

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 6342C

6 Credits: 4

Open Elective

Source-grounded

Big Data Concepts

□ To familiarize with Big Data and its sources

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 6342C is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6342C.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Artificial Intelligence & Machine Learning
Course code	6342C
Course title	Big Data Concepts
Semester	6 Credits: 4
Course category	Open Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

18 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To familiarize with Big Data and its sources
- □ To understand the basics of data preprocessing and data visualization
- □ To familiarize Big Data tools and techniques.

Course outcomes

CO	Official outcome	Study level
CO1	Familiarize the concepts of Big Data 12 Understanding	See official syllabus
CO2	Outline the basic concepts in Big Data Analytics 15 Understanding Implementation of Exploratory Data Analysis using	See official syllabus
CO3	15 Applying Python Demonstrate Map Reduce programming in Hadoop	See official syllabus
CO4	16 Applying Environment	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Familiarize the concept of Big Data	4	As specified
Module 2	Outline the basic concepts in Big Data Analytics	4	As specified
Module 3	Implement of Exploratory Data Analysis using Python	4	As specified
Module 4	Demonstrate Map Reduce programming in Hadoop Environment	6	As specified

MODULE 1

Familiarize the concept of Big Data

This module is presented from the official syllabus scope for Big Data Concepts. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Familiarize the concept of Big Data
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Familiarize the concepts of Big Data.

Familiarize the concepts of Big Data. is an official topic in Module 1 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Familiarize the concepts of Big Data. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, familiarize the concepts of big data. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Familiarize the concepts of Big Data. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Familiarize the concepts of Big Data.
 പരമ്പരാഗതമായി നിർവ്വചന/അർത്ഥം നിർവ്വചിക്കുക. നിർവ്വചനം നിർവ്വചിക്കുക
 നിർവ്വചനം, steps, application നിർവ്വചിക്കുക നിർവ്വചിക്കുക നിർവ്വചിക്കുക. Technical terms
 English- നിർവ്വചിക്കുക നിർവ്വചിക്കുക.

▼ Describe Big data Analytics and its applications

Describe Big data Analytics and its applications is an official topic in Module 1 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe Big data Analytics and its applications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe big data analytics and its applications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe Big data Analytics and its applications means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Describe Big data Analytics and its applications
 പരീക്ഷയിൽ എഴുതേണ്ടത് definition/meaning ഉൾപ്പെടെ.
 പരീക്ഷയിൽ എഴുതേണ്ടത്, steps, application ഉൾപ്പെടെ.
 Technical terms English-ൽ എഴുതേണ്ടത്.

▼ Overview of techniques used in Data Science

Overview of techniques used in Data Science is an official topic in Module 1 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Overview of techniques used in Data Science using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, overview of techniques used in data science should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Overview of techniques used in Data Science means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Overview of techniques used in Data Science
 definition/meaning .
 , steps, application . Technical terms English- .

▼ **Compare Big Data and Data Warehouse 3 Understanding Introduction to Bigdata : Data -Big data - concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse.**

Compare Big Data and Data Warehouse 3 Understanding Introduction to Bigdata : Data -Big data - concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. is an official topic in Module 1 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare Big Data and Data Warehouse 3 Understanding Introduction to Bigdata : Data -Big data - concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured - Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and

clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare big data and data warehouse 3 understanding introduction to bigdata : data -big data – concepts, characteristics (4 vs), types of digital data - structured, unstructured and semi structured -examples of big data. data science/ big data analytics : definition, requirement of high performance analytics, benefits, applications and challenges. techniques used in data science - classification, regression and clustering. data warehouse: definition, coexistence of big data and data warehouse. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare Big Data and Data Warehouse 3 Understanding Introduction to Bigdata : Data -Big data – concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Compare Big Data and Data Warehouse
 3 Understanding Introduction to Bigdata : Data -Big data - concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse.
 definition/meaning .
 steps, application
 Technical terms English-

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Outline the basic concepts in Big Data Analytics

This module is presented from the official syllabus scope for Big Data Concepts. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Outline the basic concepts in Big Data Analytics
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Summarize KDD Process.

Summarize KDD Process. is an official topic in Module 2 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize KDD Process. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize kdd process. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize KDD Process. means in the official course context
Study lens	Principle → components or stages → result → application

▼ Explain data mining techniques

Explain data mining techniques is an official topic in Module 2 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain data mining techniques using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain data mining techniques should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain data mining techniques means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Explain data mining techniques
 പരീക്ഷയ്ക്കായി definition/meaning നൽകുക. പരീക്ഷയിൽ
 പരീക്ഷിക്കുക, steps, application നൽകുക. Technical terms
 English-ൽ നൽകുക.

▼ Describe data preprocessing methods

Describe data preprocessing methods is an official topic in Module 2 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe data preprocessing methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe data preprocessing methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe data preprocessing methods means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
 Define data preprocessing methods.
 definition/meaning.
 steps, application.
 Technical terms
 English-
 .

▼ **Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining- Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages.**

Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining- Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages. is an official topic in Module 2 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official

source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining- Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline data visualization. 3 understanding knowledge discovery in databases - kdd process, steps-data cleaning, data integration, data selection, data transformation, data mining, pattern evaluation and knowledge representation. kdd-advantages and disadvantages. data mining- introduction, data mining goals, steps to discover knowledge from data (stages of the data mining process), overview of data mining techniques - association rules, classification, prediction, clustering, regression, ann, outlier detection and genetic algorithm. knowledge representation methods data preprocessing - need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. data visualization: definition, need for visualization, advantages. should be discussed

with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining- Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining- Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process),

MODULE 3

Implement of Exploratory Data Analysis using Python

This module is presented from the official syllabus scope for Big Data Concepts. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Implement of Exploratory Data Analysis using Python
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data

Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data is an official topic in Module 3 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, familiarize the eda process. 3 understanding illustrate step-by-step exploratory data should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data definition/meaning . . . , steps, application Technical terms English-

▼ 8 Applying Analysis (EDA) using Python.

8 Applying Analysis (EDA) using Python. is an official topic in Module 3 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Applying Analysis (EDA) using Python. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 applying analysis (eda) using python. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Applying Analysis (EDA) using Python. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 8 Applying Analysis (EDA) using Python.
 പരീക്ഷയ്ക്കായി നിങ്ങൾക്ക് definition/meaning നൽകേണ്ടതാണ്. നിങ്ങൾക്ക് പരീക്ഷയ്ക്ക്
 പരീക്ഷയ്ക്ക്, steps, application നൽകേണ്ടതാണ് പരീക്ഷയ്ക്ക്. Technical terms
 English-ൽ നിങ്ങൾക്ക് നൽകേണ്ടതാണ്.

▼ Summarize the concepts of NoSQL

Summarize the concepts of NoSQL is an official topic in Module 3 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the concepts of NoSQL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the concepts of nosql should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the concepts of NoSQL means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Summarize the concepts of NoSQL
 ഓരോന്നിന്റെയും definition/meaning നൽകുക. ഓരോന്നിന്റെയും
 ഓരോന്നിന്റെയും, steps, application നൽകുക. Technical terms
 English-ൽ നൽകുക.

▼ **Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL**

Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL is an official topic in Module 3 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the concepts of newsql 2 understanding eda process: definition, importance, types-univariate, bivariate and multivariate. exploratory data analysis (eda) using python: tools and libraries for eda (e.g., python, r, jupyter notebooks) import python libraries - numpy, pandas, matplotlib, seaborn. read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. data reduction, feature engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (correlation matrix, heat maps). nosql: definition, types of nosql databases, use nosql - storing and manipulating data. newsql: characteristics of newsql, comparison of sql, nosql, and newsql should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and

manipulating data. NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL

 definition/meaning

 steps, application

 Technical terms English-

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Demonstrate Map Reduce programming in Hadoop Environment

This module is presented from the official syllabus scope for Big Data Concepts. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Demonstrate Map Reduce programming in Hadoop Environment
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ Explain the fundamentals of Hadoop

Explain the fundamentals of Hadoop is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the fundamentals of Hadoop using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the fundamentals of hadoop should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the fundamentals of Hadoop means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain the fundamentals of Hadoop
 ഫണ്ടമെന്റലുകൾ ഫണ്ടമെന്റലുകൾ definition/meaning ഫണ്ടമെന്റലുകൾ. ഫണ്ടമെന്റലുകൾ ഫണ്ടമെന്റലുകൾ
 ഫണ്ടമെന്റലുകൾ, steps, application ഫണ്ടമെന്റലുകൾ ഫണ്ടമെന്റലുകൾ ഫണ്ടമെന്റലുകൾ. Technical terms
 English- ഫണ്ടമെന്റലുകൾ ഫണ്ടമെന്റലുകൾ.

▼ Concepts of HDFS

Concepts of HDFS is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Concepts of HDFS using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, concepts of hdfs should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Concepts of HDFS means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
 Concepts of HDFS
 definition/meaning
 .
 , steps,
 application
 . Technical terms English-
 .

▼ List the components of Hadoop Ecosystem

List the components of Hadoop Ecosystem is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List the components of Hadoop Ecosystem using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list the components of hadoop ecosystem should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List the components of Hadoop Ecosystem means in the official course context

▼ Illustrate the concept of HDFS and MapReduce

Illustrate the concept of HDFS and MapReduce is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the concept of HDFS and MapReduce using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the concept of hdfs and mapreduce should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the concept of HDFS and MapReduce means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate the concept of HDFS and MapReduce □□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.

▼ Use the different function in MapReduce

Use the different function in MapReduce is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Use the different function in MapReduce using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, use the different function in mapreduce should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Use the different function in MapReduce means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Use the different function in MapReduce □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ **Outline the basic concepts of Hive and Pig 3 Understanding Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. Basic concepts of Hive and Pig Text / Reference T/R**

Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han,Champaign Micheline Kamber ,Jian Pei Simon Fraser "'Data Mining T2 Concepts and Techniques Third Edition" Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, — Hadoop: The Definitive Guide Third Edition, O'reilly Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers - 2012. Online Resources SI.No Website Link 1

<https://hadoop.apache.org/> 2

<https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/> 4

<https://hive.apache.org/> 5

<https://www.tutorialspoint.com/hadoop/index.htm> 6

<https://pig.apache.org/>

Outline the basic concepts of Hive and Pig 3 Understanding Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. Basic concepts of Hive and Pig Text / Reference T/R

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The Definitive Guide Third Edition, O'reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 <https://hadoop.apache.org/> 2 <https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/> 4 <https://hive.apache.org/> 5 <https://www.tutorialspoint.com/hadoop/index.htm> 6 <https://pig.apache.org/> is an official topic in Module 4 of Big Data Concepts. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the basic concepts of Hive and Pig 3
Understanding Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. Basic concepts of Hive and Pig
Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han,Champaign Micheline Kamber ,Jian Pei Simon Fraser "'Data Mining T2 Concepts and Techniques Third Edition" Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, — Hadoop: The Definitive Guide Third Edition, O'reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1

<https://hadoop.apache.org/> 2

<https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/> 4

<https://hive.apache.org/> 5 <https://www.tutorialspoint.com/hadoop/index.htm> 6

<https://pig.apache.org/> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the basic concepts of hive and pig 3 understanding hadoop: features of hadoop, key advantages of hadoop, hadoop distributions, integrated hadoop systems offered by leading market vendors, cloud-based hadoop solutions overview of hadoop ecosystems and hadoop core components, hdfs, processing data with hadoop map reduce programming: introduction, mapper, reducer, combiner, partitioner, searching, sorting. basic concepts of hive and pig text / reference t/r book title/author t1 t1 seema acharya, subhasini chellappan, "big data analytics", wiley, 2015 jiawei han, champaign micheline kamber, jian pei simon fraser "data mining t2 concepts and techniques third edition" frank j ohlhorst, —big data and analytics: turning big data into big money, r1 wiley and sas business series, 2012. r2 tom white, —hadoop: the definitive guide third edition, o'reilly media, 2012 emc education services, data science and big data analytics: discovering, r3 analyzing, visualizing and presenting data. john wiley & sons, 2015. publishers – 2012. online resources sl.no website link 1 <https://hadoop.apache.org/> 2 <https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/> 4 <https://hive.apache.org/> 5 <https://www.tutorialspoint.com/hadoop/index.htm> 6 <https://pig.apache.org/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Outline the basic concepts of Hive and Pig 3 Understanding Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. Basic concepts of Hive and Pig Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han,Champaign Micheline Kamber ,Jian Pei Simon Fraser "'Data Mining T2 Concepts and Techniques Third Edition" Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, — Hadoop: The Definitive Guide Third Edition, O'reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources SI.No Website Link 1 https://hadoop.apache.org/ 2 https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/ 4 https://hive.apache.org/ 5 https://www.tutorialspoint.com/hadoop/index.htm 6 https://pig.apache.org/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Outline the basic concepts of Hive and Pig 3 Understanding Hadoop: Features of Hadoop, Key Advantages of Hadoop, Hadoop Distributions, Integrated Hadoop Systems Offered by

Leading Market Vendors, Cloud-Based Hadoop Solutions Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop Map Reduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting. Basic concepts of Hive and Pig

Text / Reference T/R Book Title/Author T1 T1 Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015 Jiawei Han,Champaign Micheline Kamber ,Jian Pei Simon Fraser ""Data Mining T2 Concepts and Techniques Third Edition"" Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, R1 Wiley and SAS Business Series, 2012. R2 Tom White, — Hadoop: The Definitive Guide Third Edition, O'reily Media, 2012 EMC Education Services, Data Science and Big Data Analytics: Discovering, R3 Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012. Online Resources Sl.No Website Link 1 <https://hadoop.apache.org/> 2 <https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/> 4 <https://hive.apache.org/> 5 <https://www.tutorialspoint.com/hadoop/index.htm> 6 <https://pig.apache.org/> പഠിക്കുക definition/meaning പഠിക്കുക. പഠിക്കുക പഠിക്കുക, steps, application പഠിക്കുക പഠിക്കുക പഠിക്കുക. Technical terms English-പഠിക്കുക പഠിക്കുക.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain Familiarize the concepts of Big Data.. (3 marks)**

► **Define or explain Describe Big data Analytics and its applications. (3 marks)**

► **Define or explain Overview of techniques used in Data Science. (3 marks)**

► **Define or explain Compare Big Data and Data Warehouse 3**
Understanding Introduction to Bigdata : Data -Big data - concepts, characteristics (4 Vs), types of digital data - structured, unstructured and semi structured -Examples of Big data. Data Science/ Big Data Analytics : Definition, Requirement of high performance analytics, benefits, applications and challenges. Techniques used in Data Science - classification, regression and clustering. Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse.. (3 marks)

Module 2

► **Define or explain Summarize KDD Process.. (3 marks)**

► **Define or explain Explain data mining techniques. (3 marks)**

► **Define or explain Describe data preprocessing methods. (3 marks)**

► **Define or explain Outline data visualization. 3 Understanding Knowledge Discovery in Databases - KDD Process, steps-data cleaning, data integration, data selection, Data Transformation, Data Mining, Pattern Evaluation and Knowledge representation. KDD-advantages and disadvantages. Data Mining- Introduction, Data Mining Goals, Steps to discover knowledge from data (Stages of the Data Mining Process), Overview of data mining Techniques - Association Rules, Classification, Prediction, Clustering, Regression, ANN, Outlier Detection and Genetic Algorithm. Knowledge Representation Methods Data Preprocessing - Need of data preprocessing, data quality, data preprocessing methods- cleaning, integration, transformation, reduction, discretization, normalization. Data Visualization: Definition, Need for Visualization, Advantages.. (3 marks)**

Module 3

► **Define or explain Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data. (3 marks)**

► **Define or explain 8 Applying Analysis (EDA) using Python.. (3 marks)**

► **Define or explain Summarize the concepts of NoSQL. (3 marks)**

► **Define or explain Outline the concepts of NewSQL 2 Understanding EDA process: Definition, importance, types-univariate, bivariate and multivariate. Exploratory Data Analysis (EDA) using Python: Tools and libraries for EDA (e.g., Python, R, Jupyter Notebooks) Import python libraries - numpy, pandas, matplotlib, seaborn. Read and analyze datasets - reading data, analyzing, checking for duplication, and missing values calculation. Data Reduction, Feature Engineering, univariate analysis (swarmplots and countplots), bivariate analysis (pair plot, violin plot, scatter plot, box plot), multivariate analysis (Correlation Matrix, heat maps). NoSQL: Definition, Types of NoSQL Databases, Use NoSQL - Storing and manipulating data. NewSQL:**

Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL. (3 marks)

Module 4

► **Define or explain Explain the fundamentals of Hadoop. (3 marks)**

► **Define or explain Concepts of HDFS. (3 marks)**

► **Define or explain List the components of Hadoop Ecosystem. (3 marks)**

► **Define or explain Illustrate the concept of HDFS and MapReduce. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Familiarize the concepts of Big Data..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Summarize KDD Process..**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Familiarize the EDA process. 3 Understanding Illustrate step-by-step Exploratory Data.**

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Explain the fundamentals of Hadoop.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://hadoop.apache.org/>
- <https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/>
<https://hive.apache.org/>
- <https://www.tutorialspoint.com/hadoop/index.htm> <https://pig.apache.org/>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 6342C. Verify institutional updates against the current official curriculum.